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# PAPER\_1122: Bow-Shock ISM Chemistry and Prebiotic Enrichment in the UQFF Framework

## Abstract

We implement a UQFF calculator for bow-shock chemistry in the interstellar medium, based on arXiv:1808.01439 (2018). Stellar bow shocks create standoff structures where post-shock temperatures activate endothermic chemical pathways, producing OH, H<sub>2</sub>O, and SiO from grain sputtering. The bow-shock standoff distance, post-shock temperature, and chemical activation efficiencies are modeled within the UQFF  $C(t)$  molecule release framework, linking [SCm]-[UA] interactions to prebiotic enrichment.

## 1. Bow-Shock Standoff Distance

The distance from the star at which ram pressure balances stellar wind pressure:

$$R_{\text{bs}} = \sqrt{\frac{\dot{M} \cdot v_w}{4\pi\rho_{\text{ISM}}v_*^2}}$$

where  $\dot{M}$  is the mass-loss rate,  $v_w$  the wind velocity,  $\rho_{\text{ISM}}$  the ambient density, and  $v_*$  the stellar space velocity.

## 2. Post-Shock Temperature

From the Rankine-Hugoniot strong-shock conditions:

$$T_{\text{ps}} = \frac{3\mu m_p v_s^2}{16k_B}$$

For  $v_* = 30$  km/s and  $\mu = 1.27$ :  $T_{\text{ps}} \approx 5700$  K.

## 3. Chemical Activation

Endothermic barriers determine which molecules form:

| Molecule         | $T_{\text{crit}}$ (K) | Mechanism                                                           |
|------------------|-----------------------|---------------------------------------------------------------------|
| OH               | 2000                  | Gas-phase: $\text{O} + \text{H}_2 \rightarrow \text{OH} + \text{H}$ |
| H <sub>2</sub> O | 1500                  | Grain surface catalysis                                             |
| SiO              | 3500                  | Grain core sputtering                                               |

Activation efficiency:  $\eta = \sigma(T_{\text{ps}} - T_{\text{crit}})$  (sigmoid function).

## 4. UQFF Alignment

The bow-shock C(t) enrichment pathway supports [SCm]-[UA] prebiotic chemistry at **80% alignment** with observational data. Cooling timescales determine the chemical freeze-out composition.

## References

- arXiv:1808.01439 — Bow-shock chemistry in the ISM (2018).
- Ceccarelli & Codella (2024) — Shock-triggered star formation chemistry.